Robayed Ashraf

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SUMMARY

AI/ML Engineer specialising in building production-grade systems across agentic RAG pipelines, LLM applications, and real-time computer vision. Built and deployed DocuMind end-to-end, a self-correcting, agentic RAG system with hybrid search, hallucination detection, and RAGAS evaluation. Comfortable owning the full stack from model design and API development to containerisation and CI/CD, and driven by shipping things that work in the real world.

TECHNICAL SKILLS

AI & Machine Learning: Agentic AI, LLM Applications, RAG, Generative AI, Deep Learning, Computer Vision, NLP, Prompt Engineering, Hybrid Search, Vector Embeddings, LLM Evaluation (RAGAS)
Frameworks: Python, PyTorch, TensorFlow, Keras, LangChain, LangGraph, FastAPI, OpenCV, Scikit-Learn, HuggingFace, ChromaDB
MLOps & DevOps: Docker, Docker Compose, GitHub Actions (CI/CD), ClearML, Pytest, REST API Development
Tools & Practices: Git, SQL, Agile/Scrum, Jira, Confluence

PROJECTS

DocuMind: Agentic Document Intelligence Feb 26 – Present
Tech: LangGraph, LangChain, Gemini 2.5 Flash, FastAPI, ChromaDB, HuggingFace, BM25, Docker, GitHub Actions, RAGAS
- Built a production-grade agentic RAG system with autonomous query routing, hallucination detection, and self-correcting retry loops using LangGraph and Gemini 2.5 Flash.
- Engineered hybrid retrieval combining BM25 and dense vector embeddings, fused via Reciprocal Rank Fusion with cross-encoder reranking for enterprise-grade retrieval precision.
- Deployed as a fully containerised multi-service application with SSE streaming, per-session memory, and a REST API supporting upload, indexing, querying, and streaming.
- Shipped with a RAGAS evaluation suite measuring faithfulness, answer relevancy, and context quality across pipeline iterations.

SignSync: Sign Language Translation for Video Conferencing Feb 25 – Jun 25
Tech: Python, ANN, OpenCV, Gemini API (LLM integration), Computer Vision
- Trained an ANN on 3,200 images across 42 classes for real-time AUSLAN finger spelling classification, achieving 99.79% validation accuracy.
- Integrated Gemini API for LLM-based post-processing, correcting recognition errors, and improving caption readability in real time.
- Designed a 30-frame confidence-based smoothing mechanism to stabilise live predictions in continuous video streams.
- Deployed the full pipeline into a live Jitsi environment, delivering real-time sign-to-text captions as a production-viable accessibility feature.
- Presented at UTS Tech Fest AI Showcase 2025, recognised for practical impact among postgraduate AI projects.

ThirdAxis: Smart Plant Health Monitor Feb 25 – Jun 25
Tech: Python, CNN, ClearML, Streamlit, GitHub Actions, Jira, Confluence
- Managed end-to-end ML project lifecycle using Agile practices with Jira and Confluence, covering data ingestion, preprocessing, training, evaluation, and deployment.
- Designed a modular ClearML pipeline with separate scripts for each stage, enabling reproducible and traceable experimentation.
- Implemented ClearML artifact management for dataset and model versioning, supporting experiment comparison and rollback across training iterations.
- Built a Streamlit inference interface for real-time plant disease classification from leaf images, addressing early disease detection needs in precision agriculture.

Multi-Object Tracking System for Real-Time Video Inference Jan 25 – Jun 25
Tech: PyTorch, OpenCV, YOLOX, MOTR, TrackEval, CUDA, Streamlit

- Designed and evaluated a real-time live video tracking system, comparing MOTR (Transformer-based) and BoostTrack++ (tracking-by-detection with YOLOX) to assess feasibility for retail footfall analytics and smart city surveillance.
- Profiled inference on NVIDIA RTX 4060 (CUDA/PyTorch), recording 1.54 to 3.60 FPS and 255 to 409ms latency to identify deployment trade-offs for live video scenarios.
- Built a modular Streamlit prototype supporting webcam and live video inference for real-time visual comparison of tracking outputs across both architectures.

WORK EXPERIENCE

Self-Employed, AI/ML Engineer Feb 26 – Present

- Designed the full agentic pipeline architecture from scratch, making deliberate choices around query routing, relevance grading, and hallucination detection.
- Improved retrieval precision by combining BM25 with dense vector search and cross-encoder reranking, iterating until results were meaningfully better than a vector-only baseline.
- Owned the evaluation strategy end-to-end, using RAGAS metrics to identify and fix quality gaps across pipeline iterations.
- Built the complete deployment setup, including containerisation, CI/CD, and REST API, taking the project from prototype to a shippable system.

East West University Computer Programming Club, Programming Trainer (Volunteering) Jan 19 – Oct 21

- Mentored 50+ students in algorithms, data structures, and competitive programming, strengthening their analytical and problem-solving foundations relevant to AI and ML engineering.
- Designed and conducted ICPC-style mock contests to simulate high-pressure, time-critical programming environments, developing structured evaluation and technical communication skills.

EDUCATION

University of Technology Sydney, Master of Artificial Intelligence (Major: Computer Vision) Aug 23 – Jun 25
WAM: 80.94 / 100 (Grade: 6.13 / 7.00)

East West University, Bachelor of Science in Computer Science and Engineering Jan 18 – Oct 21
CGPA: 3.64 / 4.00

- Dean's List recipient and merit-based scholarship awardee for academic excellence.